

**From beginner to low-level programmer:
learn how real Linux software is built.**

Before mastering Linux system programming, you must understand the tools used to build, debug, analyze, and study software.

This introductory volume presents the Linux development environment through practical use of GCC, Make, Autoconf, GDB, documentation systems, linking, compilation, and debugging techniques. It also introduces standards, libraries, and the structure of Linux software development.

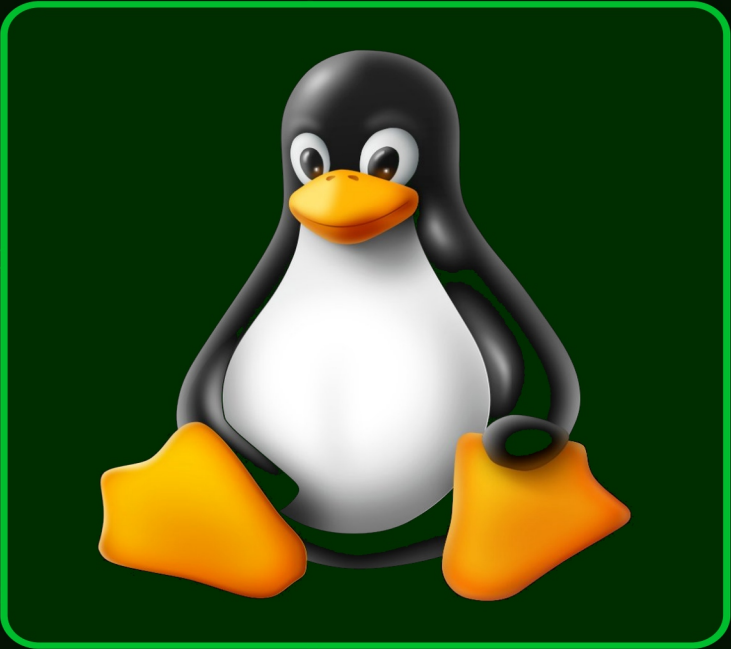
Written for beginners and aspiring low-level programmers, this book provides the foundation needed to read, build, analyze, and truly understand system-level code.

```
root@linux:~# ./advanced_programming.sh
```

Advanced Programming in the Linux Environment

Volume 0

Tools



```
#!/bin/bash
#include <stdio.h>
int main(void) {
    fork(); execve{};
    return EXIT_SUCCESS;
}
```

```
$ gcc -o prog main.c
$ ./prog
Hello, Linux!
$ make install
$ sudo !!
```